

Accessibility Evaluation Through Heat Maps: ECCR and LEEDS

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Why Accessibility is Important

Bridging Information Gaps

- This project provides a "from students, for students" database so individuals can understand a building's layout before they arrive.

Addressing Systemic Barriers

- Survey data reveals that 1 in 5 buildings on campus have sections that make entering or exiting difficult for students with disabilities

Closing Support Gaps

- Currently, 1 in 3 students report finding at least one floor in every campus building with little to no accessibility support.

Project Objective

- The goal of our project was to create a database that could help the university in understanding how accessible there campus is from students for students
- This is to help students to be able to look at a building and be able to understand if they can or cannot get around that building on campus to hopefully be able to accommodate students with disabilities

Data Collection + Methods

- Starting of with a survey to students, this was our first step in the process to find which building are a problematic or have a lack of accessibility that students feel like.
- On site research, we conducted a survey of the **Engineering Center** as well as the **Leeds School of Business** building. From these surveys we looked at each of the floors and based off of our grading we were able to determine whether a floor was accessible or not for students

Survey Responses

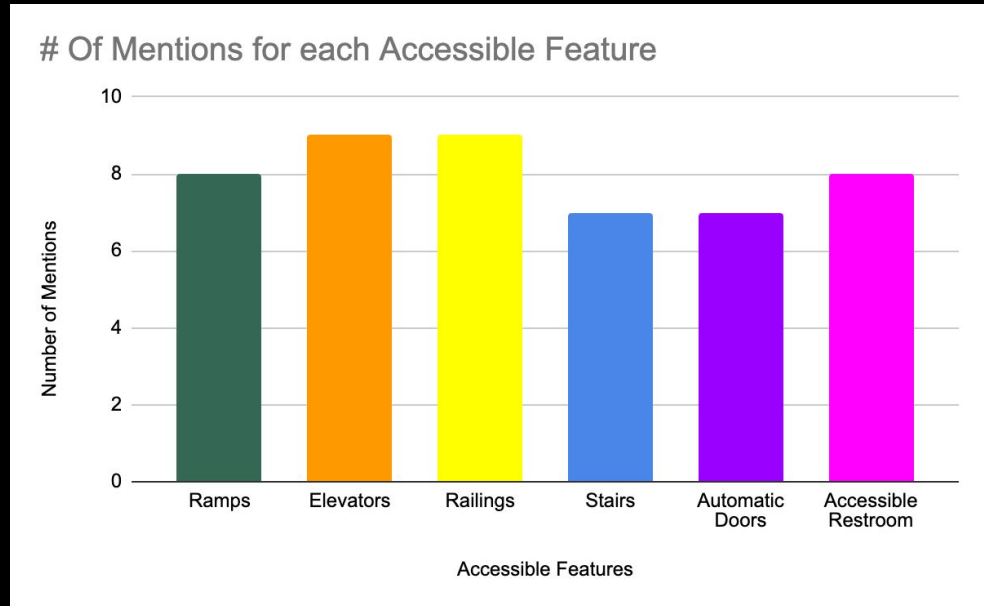
- From our surveys we found that almost 1 in 5 buildings on campus had parts to them that made getting in or even out of the building difficult for students who have a disability.
- As well we found that almost 1 in 3 students found that at least 1 floor of each building on campus had little to no support for students with disabilities

Survey Responses p.2

Many of our survey respondents had mixed feelings about the engineering building.

When we asked how the Engineering Building could be improved:

- "I feel the signage across the Engineering could be improved a lot as its very confusing for almost everyone that walks into the building"
- "Too many sharp corners that cause major traffic problems and accidents"



Accessibility Scoring Rubric

0-5 Normalized Scale

- Scores are calculated via an additive formula and subjected to a 5-point range to match heat map visualizations.
- **5** = Highly Accessible; **0** = Inaccessible or Blocked.

Weighted Point Breakdown

- **Route Access (0 to +2)**: Evaluates path width and lack of obstructions for mobility devices.
- **Entry & Doors (-1 to +1)**: Scored on level transitions and automatic vs. heavy/narrow manual doors.
- **Space & Usability (-1 to +1)**: Measures turning room and the functional usability of desks, labs, or restrooms.

Critical Failures (Automatic Score of 0)

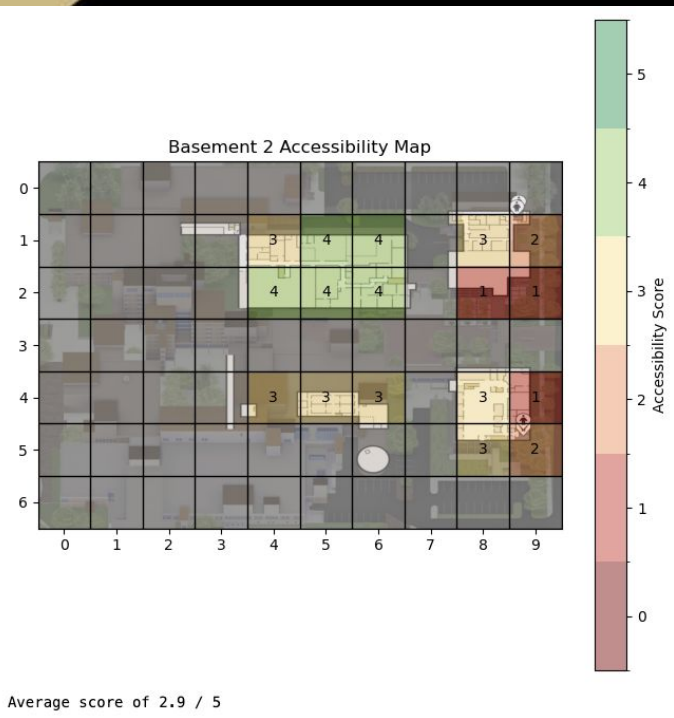
- Any cell containing stair-only access or an inaccessible entrance is forced to a score of 0.
- Major route blockages or non-functional automatic doors become maximum point deductions.

Context-Specific Evaluation

- Grading criteria shift based on cell type: **Circulation** (hallways), **Threshold** (entryways), or **Destination** (classrooms/labs).

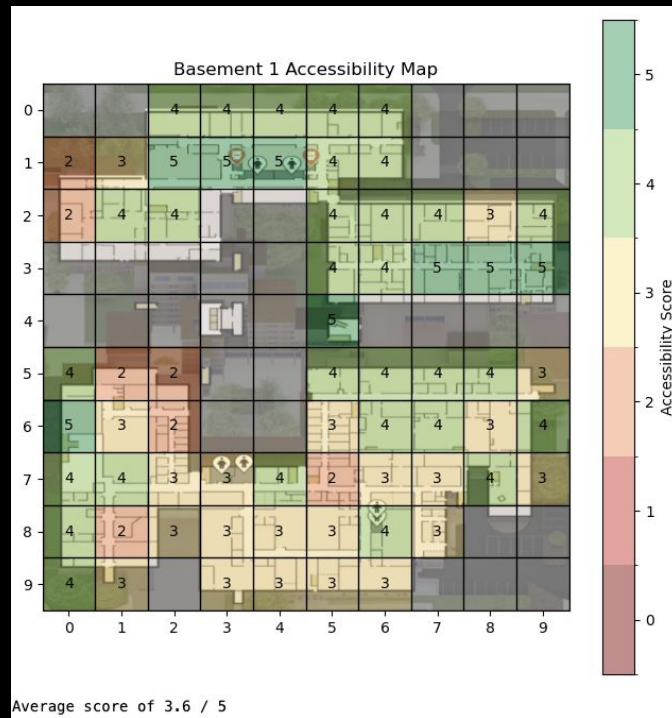
Visualizations ECCR

Basement 2 ECCR



Average score = 2.9 / 5

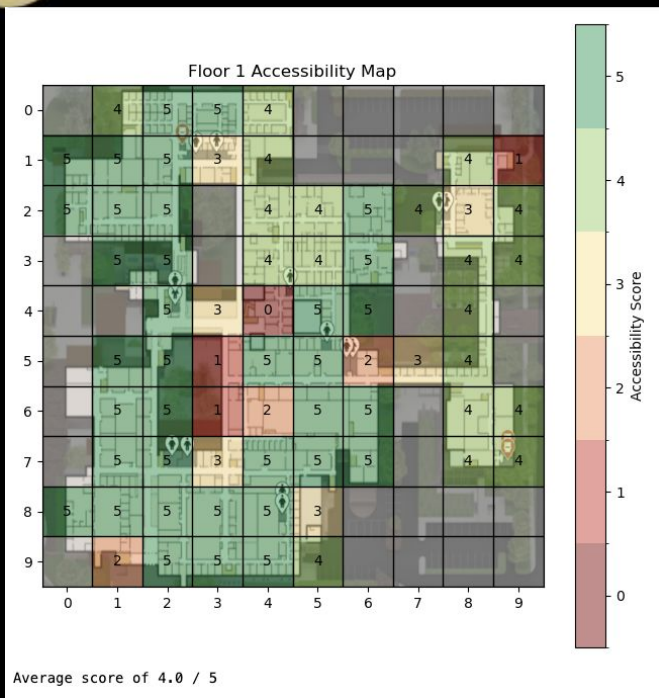
Basement 1 ECCR



Average score = 3.6 / 5

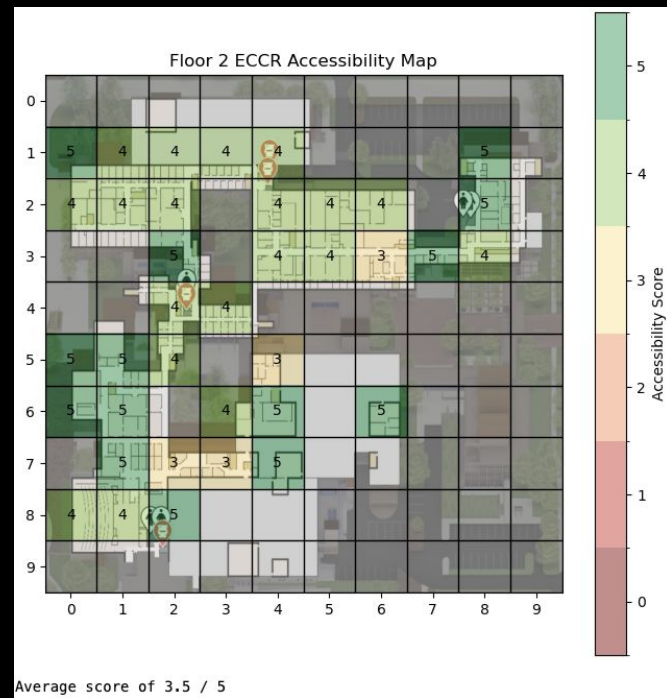
Visualizations ECCR

Floor 1 ECCR



Average score = 4.0 / 5

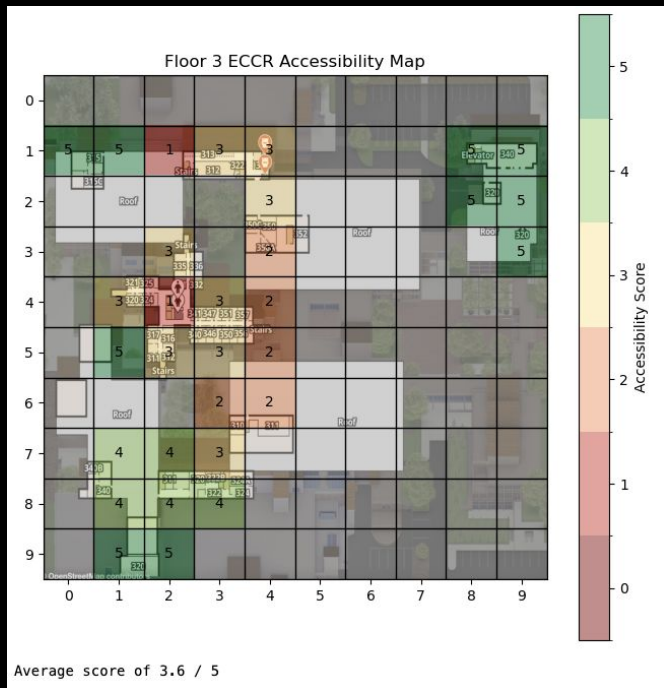
Floor 2 ECCR



Average score = 3.5 / 5

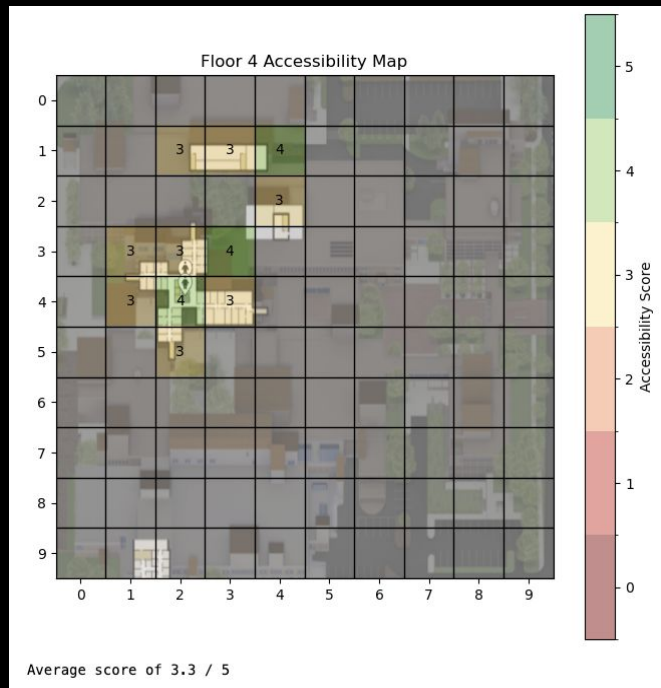
Visualizations ECCR

Floor 3 ECCR



Average score = 3.6 / 5

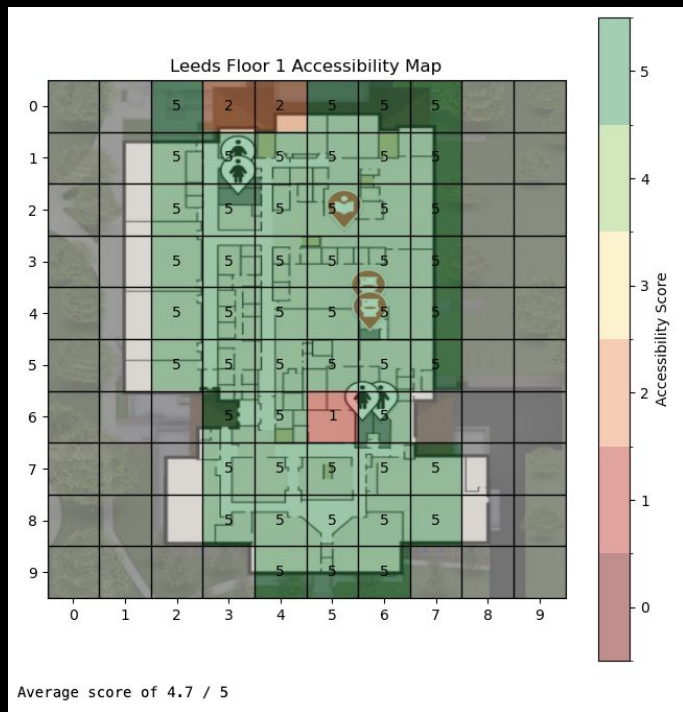
Floor 4 ECCR



Average score = 3.3 / 5

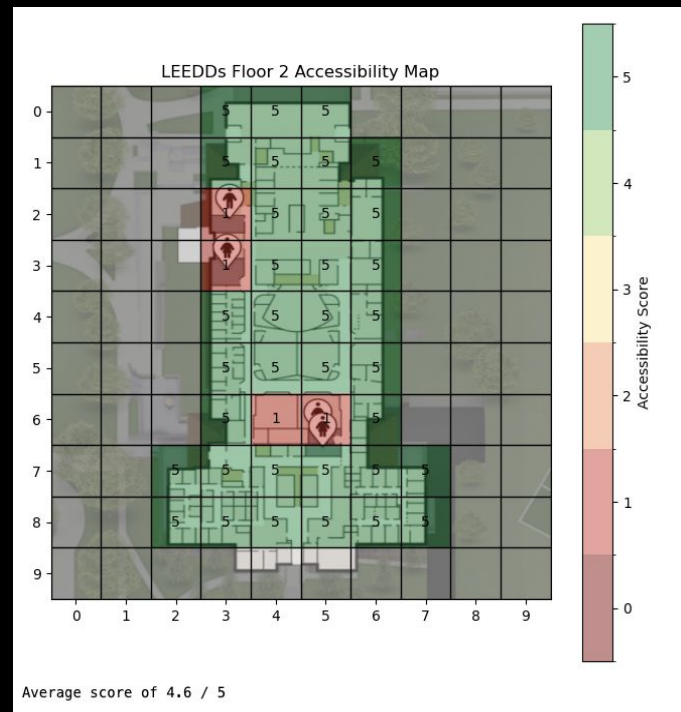
Visualizations LEEDs

Floor 1 LEEDs



Average score = 4.7 / 5

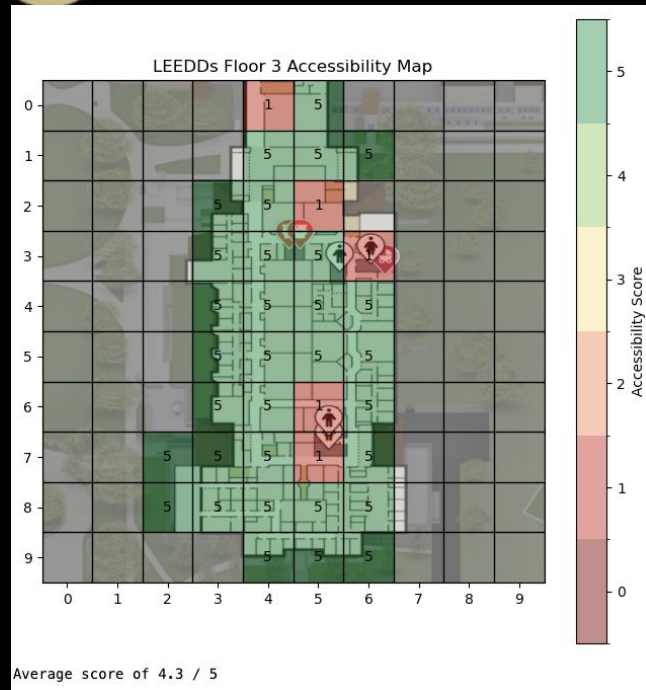
Floor 2 LEEDs



Average score = 4.6 / 5

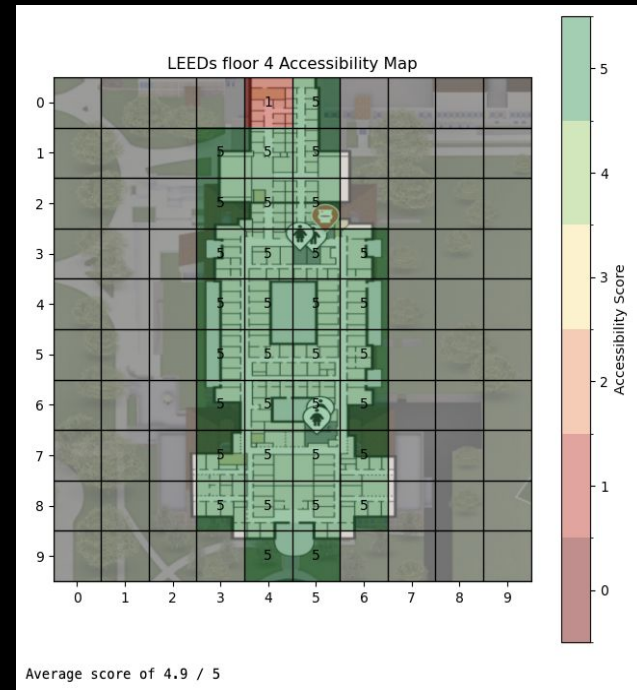
Visualizations LEEDs

Floor 3 LEEDS



Average score = 4.3 / 5

Floor 4 LEEDS



Average score = 4.9 / 5

ECCR Findings

Based on our research, we found that the ECCR building could improve in several ways

Issues:

- Confusing for virtually anyone, the layout is not uniform
- Alternatives to staircases are difficult to find
- Signage is not usable for those with visual impairments
- Accessible bathrooms are only located on floor 1 and 2



LEEDs Findings

Based on our research, we found that the Leeds building was mostly satisfactory

Negatives:

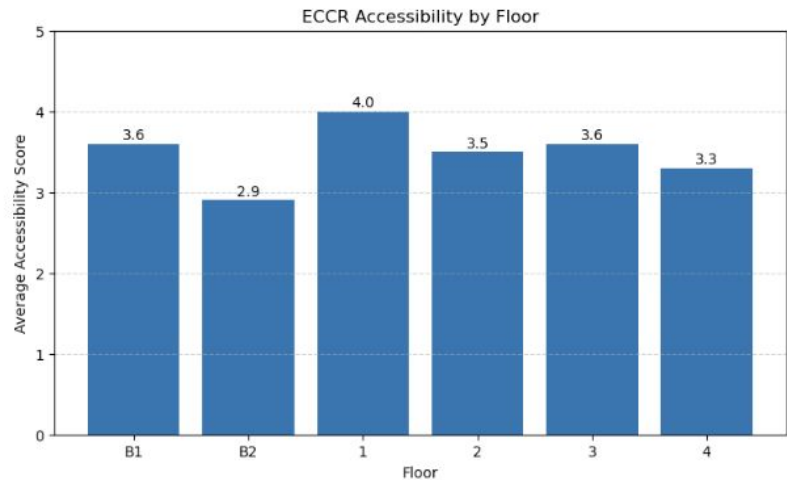
- There are several large staircases, without clear signage as to where alternatives may be
- Classroom doors are relatively heavy and can only be opened by the handle

Positives:

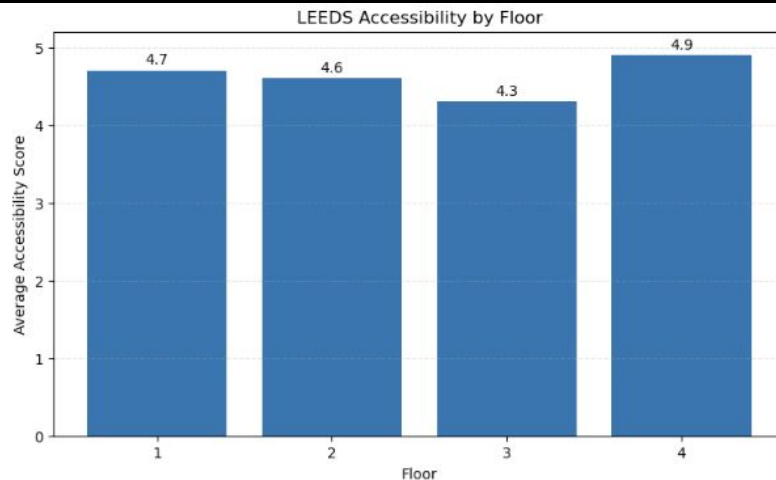
- There are several elevators that make each floor accessible
- There are also ramps near smaller staircases although they may require a longer route for certain classrooms.

ECCR vs. LEEDs Comparison

Average Building Score = 3.5 / 5



Average Building Score = 4.63 / 5



- LEEDs showed a higher overall average accessibility score than ECCR (4.63 vs. 3.5), suggesting a more consistently accessible building layout.
- ECCR varied more by floor, especially in the basement levels.

Recommendations

ECCR:

- Many of its accessibility issues are unfortunately innate to the buildings design
- Better signage for those with visual impairments, perhaps in braille, route recommendations

Leeds:

- Classroom doors could be left propped open at least between classes
- Signage towards elevators and accessibility ramps
-

Limitations

Subjectivity of Evaluation: Grid scores represent "survey units" based on student observations rather than professional ADA compliance audits.

Grid-Scale Constraints: The use of a **10x10** grid for heat maps means specific physical nuances may be missed or averaged out within a single cell.

Limited Building Scope: The research focused specifically on the **ECCR** and **Koelbel**; these layouts may not reflect accessibility challenges of differently designed campus buildings.

Contextual Grading: Scores in "N/A" areas (exterior landscaping or non-essential zones) were omitted to focus on high-traffic student areas

Conclusion

We found that Improvements could certainly be made for the buildings we researched. Some of this improvements would be fairly straightforward, while others involve critical design flaws that cannot realistically be changed. We believe that our research could help inform future improvements to the accessibility of these buildings.

Visualization Code

```

1 # Visualizations for floors 2 of the ECCR
2
3
4 import numpy as np
5 import matplotlib.pyplot as plt
6 import matplotlib.image as mpimg
7 from matplotlib.colors import ListedColormap, BoundaryNorm
8
9 # Loading the floor image
10 img = mpimg.imread("Screenshot 2026-04-04 at 4.22.33 PM.png")
11
12 # 10x10 score matrix
13 # Use np.nan for areas that are being evaluated not evaluating
14 scores = np.array([
15     [np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan],
16     [5, 4, 4, 4, np.nan, np.nan, np.nan, 5, np.nan],
17     [4, 4, 4, np.nan, 4, 4, 4, np.nan, 5, np.nan],
18     [np.nan, np.nan, 5, np.nan, 4, 4, 3, 5, 4, np.nan],
19     [np.nan, np.nan, 4, 4, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan],
20     [5, 5, 4, np.nan, 3, np.nan, np.nan, np.nan, np.nan, np.nan],
21     [5, 5, np.nan, 4, 5, np.nan, 5, np.nan, np.nan, np.nan],
22     [np.nan, 5, 3, 3, 5, np.nan, np.nan, np.nan, np.nan, np.nan],
23     [4, 4, 5, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan],
24     [np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan, np.nan]
25 ])
26
27 # colors for score from 0 - 5
28 colors = [
29     "#8B0000", # 0 dark red
30     "#D73027", # 1 red
31     "#FC8D59", # 2 orange
32     "#FFEB3B", # 3 yellow
33     "#91CF60", # 4 light green
34     "#1A9850", # 5 green
35 ]
36
37 cmap = ListedColormap(colors)
38 # code below for np.nan cells
39 cmap.set_bad(color="lightgray")
40
41 # Boundaries for bins within the 10 x 10 grid
42 bounds = [-0.5, 0.5, 1.5, 2.5, 3.5, 4.5, 5.5]
43 norm = BoundaryNorm(bounds, cmap.N)
44
45 fig, ax = plt.subplots(figsize=(8, 8))
46

```

```

# Put the floor of the ECCR plan image in the background
ax.imshow(img, extent=[-0.5, 9.5, 9.5, -0.5], aspect="equal", zorder=0)

# Put the heatmap on top of the ECCR image
im = ax.imshow(scores, cmap=cmap, norm=norm, alpha=0.45, zorder=1)

# Grid lines
ax.set_xticks(np.arange(-0.5, 10, 1), minor=True)
ax.set_yticks(np.arange(-0.5, 10, 1), minor=True)
ax.grid(which="minor", color="black", linestyle="-", linewidth=1)
ax.tick_params(which="minor", bottom=False, left=False)

# set ticks
ax.set_xticks(np.arange(10))
ax.set_yticks(np.arange(10))

# Adding score labels inside each tile
for i in range(scores.shape[0]):
    for j in range(scores.shape[1]):
        if not np.isnan(scores[i, j]):
            ax.text(j, i, int(scores[i, j]),
                    ha="center", va="center", color="black", fontsize=10, zorder=2)

# Colorbar on side of visualization
cbar = plt.colorbar(im, ax=ax, ticks=[0, 1, 2, 3, 4, 5])
cbar.set_label("Accessibility Score")

ax.set_title("Floor 2 ECCR Accessibility Map")
plt.show()
print("Average score of 3.5 / 5")

```